

HUMANOID FALL AVOIDANCE USING A MIXTURE OF STRATEGIES

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If we are to one day rely on robots as assistive devices they should be capable of mitigating the impact of random disturbances and avoid falling. Humans are surprisingly apt at remaining on their feet when pushed; they rely on reflexes such as bending the ankles and/or the hips, or by taking a step if the magnitude of the disturbance is relatively large. This paper presents a fall avoidance scheme that is capable of applying both ankle and hip strategies on a humanoid robot. While both strategies serve the same purpose, the hip strategy can absorb larger disturbances but has a higher energy overhead and should be avoided when it is not necessary. Our system is capable of detecting at the onset of a disturbance if an ankle or hip strategy is more appropriate. The decision is taken based on a 'decision surface' that is delimited by threshold values of the robot's state variables. The control is based on the Virtual Model Control (VMC) approach. The system is tested on a simulated robot developed under Gazebo as well as on a real small-scale humanoid robot. Results show successful fall avoidance with an ability to choose the optimum fall avoidance strategy.

Keywords: Fall avoidance, ankle strategy, hip strategy.

1. Introduction

For humanoids to be more human-like they must learn among other things how to walk, run, stand up, and avoid falling. Fall avoidance is an especially important capability to possess. One just has to watch a RoboCup match in the Humanoid league to appreciate how valuable such an ability might be. During these games robots are frequently bumped and fall to the floor. Although they are capable of

getting up again, time is wasted and therefore puts their team at a disadvantage. Fall avoidance benefits are not limited to winning a football match. When a robot falls there is a risk that it might damage itself from the fall, which might lead to financial losses and downtime for servicing.

Human beings deal with disturbances via their reflexes and take corrective actions such as bending the ankles, bending the hips while rotating the arms, or taking lunge steps in the case the disturbance is excessive.² Emulating such reactions in humanoid robots is difficult because of their naturally unstable nonlinear multivariate dynamics and their limited foot-ground interaction.³ Nevertheless, by posing constraints and simplifications to the problem, where only the characteristics of interest are preserved and the rest are discarded, major contributions to humanoid stability have been possible.

When a human is pushed with a relatively low force his/her instinctive reflex is to counteract this disturbance with a torque at the ankles. This strategy is known in the literature as the *ankle strategy*. Although effective at relatively low disturbances, a more effective countermeasure is required if the disturbance is larger. One strategy used by humans is to create a momentum by rotating the torso at the hips; a strategy known as the *hip strategy* or the *momentum strategy*. If the push is yet larger and both the ankle and hip reflexes are not sufficient, then the human is forced to take a step in order not to fall. All three of these strategies have been studied in the literature^{4–7} but to our knowledge no system exists today that like humans can assess at the time of impact of the disturbance which of the above strategies should be taken such that the robot consumes the least amount of energy — ankle method being the least energy consumer followed by the hip and stepping methods — and yet avoids falling.

Hyon *et al.*²¹ implement a controller that acts to keep the Center of Mass (CoM) at a relatively fixed location. The advantage of the system is that it is capable of recovering from a disturbance and avoid falling down when pushed from the front, back, and side. The authors also implemented a stepping strategy in the case the reaction force underneath a foot becomes zero. The disadvantage in the system that they propose is that it does not include any momentum controller. In fact, as shown in our results as well as in other papers^{7,14} momentum can be used to recover from disturbances that otherwise cause the robot to fall down. Another shortcoming in Hyon *et al.*'s work is that the trigger for stepping is initiated for a special condition of having the value of the reaction force underneath a certain foot equal to zero. This is not a generic condition, and better results could be obtained if the robot could realize at the onset of the disturbance if stepping is required.

Nenchev *et al.*^{20,19} apply the ankle and hip strategies on a humanoid robot in order to recover from both an impulse and/or a continuous disturbance. They develop what they refer to as the *reaction null space* of a humanoid, which allows the system to apply a hip strategy combined with an ankle rotation in order to prevent any rotation of the foot. Two shortcomings are identified here. First, the imposed foot rotation constraint could be misleading to the humanoid in terms of identifying

instability given that the robot could still recover from a disturbance while standing on its tip-toes. In our work this constraint is removed and Virtual Model Control is used in such a way to combine both ankle and hip strategies with the objective of regulating the position of the CoM. Our results show the robot recovering even after the foot starts rotating. Second, the criterion by which the system decides the optimum fall avoidance strategy is ad-hoc and based on a hard body acceleration threshold. The question here is what would occur if a disturbance is applied while the robot is not at its equilibrium position? Nenchev's humanoid would use the same threshold instead of one that is dependent on the current robot stance; a risk that could lead to taking a wrong strategy and falling down. An alternative solution is to consider an analytical approach such as a decision surface that is capable of informing the system of the robots instantaneous state; thereby allowing a more educated choice regarding which fall avoidance strategy is more appropriate.

In light of the progress achieved above in humanoid fall avoidance and the shortcomings that are highlighted in each approach, this paper presents a fall avoidance scheme with effective control and software scheduling capable of deciding when each strategy is more appropriately applied. The system is based on developing what is known as a *decision surface*, which is a graph showing the limits of the states (angular position and angular velocity) beyond which each of the aforementioned strategies is no longer effective at fall avoidance, thereby warranting a more aggressive and power hungry strategy. While this decision surface was developed previously,⁷ in our work these surfaces are generalized to be used on a bipedal robot model instead of only on an inverted pendulum and flywheel models. Another design constraint that is posed in our system is that it is implemented on a humanoid that processes limited processing capabilities. In this context, the control system is simplified from that of a nonlinear one to a simpler and more intuitive controller using a Virtual Model Control (VMC) system. VMC considerably simplifies the control of the robot during fall avoidance by modeling the control system via lumped springs and dampers. VMC is used to allow the combination of hip and ankle strategy in a single fall avoidance scheme, thus using momentum compensation as well as CoM balancing when ankle strategy on its own is not sufficient to recover from a disturbance. Finally, in the models that are used in this paper the degree of freedom of the robot at the location of the foot-ground interaction is differentiated from that of the ankle, hence foot rotation is not considered an issue in our system but on the contrary takes parts of the decision surface development as well as the control strategies.

In our previous work¹⁴ momentum strategy alone was realized and implemented on both a simulated robot (named BAUB) as well as on a small humanoid (Bioloid robot)¹⁰ that we retrofitted with additional sensors for the purpose of fall avoidance. In this paper the integrated hip/ankle system is implemented on BAUB as well as Bioloid humanoid robot.

The remainder of this paper is structured as follows. Section 2 covers the background on fall avoidance, where a critical review of the state of the art is presented. Section 3 introduces the control system upon which our system is based. Section 4

discusses the application of fall avoidance on a 3-DoF humanoid model. Section 5 presents the results of the fall avoidance scheme applied on our simulated and real humanoid. Finally Sec. 6 discusses the limitations of our system then concludes the paper with the contributions of our work and the direction of our future work.

2. Fall Avoidance

A bipedal robot is inherently unstable by nature. In its simplest form, all the humanoid's limbs and joints situated above the ankle joint are considered to be stiff and the humanoid is modeled as an inverted pendulum. This model is valid as long as the foot is flat over the ground and the ankle joint can be controlled as one would control a joint of a robotic arm, albeit with the added constraint of the limited foot-ground interaction.³ As soon as the disturbance on the robot exceeds a certain magnitude, its foot starts rotating and the robot runs the risk of losing controllability of its foot-ground degree of freedom. This does not necessarily mean that the robot is unstable from the classical control point of view since during a fall it can always use its upper body dynamics to compensate (*e.g.* taking a step, rotating arms, or leaning on static objects).

In the vocabulary of humanoid dynamics, three definitions are used to describe the mechanics of foot-ground interaction. These definitions include the Center of Pressure (CoP), the Zero Moment Point (ZMP) and the Foot Rotation Indicator (FRI). CoP is defined as the location where an equivalent ground reaction force under the foot's support is represented. The force at this position is the integral of the normal pressure distribution under the foot. The CoP can be measured using force sensors placed under the foot. According to this definition, it makes intuitive sense that the CoP can never leave the support polygon under the foot soles.

ZMP¹⁵ on the other hand is the location on the ground where the robot's inertial, gravitational, Coriolis, and centrifugal forces are balanced by the reaction force. This can be translated to indicate that the ZMP is the position of the CoP that would cancel any moment due to inertia and gravity. Goswami,⁴ proved that this definition of the ZMP indicates that the ZMP is in fact the CoP. The ZMP can only exist if it were equal to the CoP, else the gravity and inertia are not compensated and there is no zero moment point according to the previous definition of the term. In static equilibrium, the CoP (or ZMP) is situated inside the support polygon, but in reality, CoP can reach the boundary of this polygon and still be in equilibrium. Since the position of the CoP cannot distinguish between the marginal state of static equilibrium and a complete loss of equilibrium, where in both cases it is situated at the support boundary, its utility in stability analysis is limited. The main reason for this limitation is that the location of CoP can never reach beyond the extents of the foot soles.

In light of this shortcoming, Goswami put forward the concept of the FRI point, also called the Fictitious ZMP (FZMP), which is the point on the foot/ground contact surface, within or outside the convex hull of the foot-support area at which the resultant moment of the force/torque impressed on the foot is normal to the

surface. By that definition, the FRI point on the ground can be understood as the location on the ground where the CoP is desired in order for the robot to be in dynamic equilibrium. Therefore, if the FRI point is inside the foot support then FRI is equal to the CoP but if FRI is outside of the foot support it becomes a fictitious point, which can act as an indicator of the degree of stability of the robot.

In stance balancing push recovery, the stability problem of a bipedal robot consists of actually keeping the robot angle in a bounded region in space. Whether the CoP reaches the edge of the foot or not is of no importance in this case. What is important is that during a push the robot has the ability to maintain controllability of the foot-ground interaction. This can be achieved by using strategies inspired by the biomechanics of human reflex such as the ankle strategy, the momentum strategy, and the stepping strategy.

2.1. Ankle strategy

In this mode a human compensates for a disturbance just like an inverted pendulum does, simply by providing an ankle torque that counteracts the disturbance and prevents the fall. Due to the limited foot ground interaction, the ankle torque is limited and can only be used as a preventive falling measure in the case of small disturbances.

The inverted pendulum model of a biped was first pointed out by Hemami *et al.*¹⁶ The body as a whole is treated as a point mass located at the CoM and only the muscle dynamics at the ankle joint are involved (see Fig. 1). The equation of motion for this model is

$$mL^2\ddot{\theta} = mgL \sin \theta + T_a, \quad (1)$$

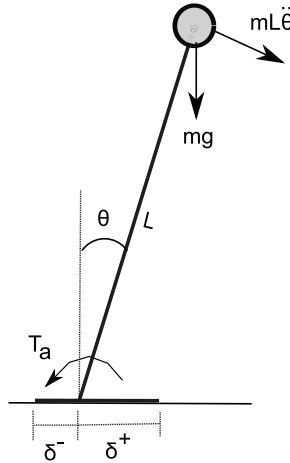


Fig. 1. The inverted pendulum model.

where T_a is the torque input at the ankle joint, m , g , L are respectively the mass, gravity and distance from the ankle to the CoM. By linearizing the gravity term for small θ it can be considered that $\sin(\theta) \simeq \theta$ and (1) becomes

$$\ddot{\theta} = \omega^2 \theta + \frac{T_a}{mL^2}, \quad (2)$$

where $\omega = \sqrt{\frac{g}{L}}$.

Deciding when to apply which fall avoidance strategy is based on a model that is developed for each strategy, which is capable of inferring from the current state (e.g., angular position and angular velocity) of the humanoid if it can recover by applying the corresponding strategy. Stephens⁷ was the first to introduce what is known as decision surfaces for humanoid push recovery strategies. A decision surface delineates the thresholds (or 2D region if we consider a 2DoF robot) beyond which the robot can not avoid falling by applying the corresponding fall avoidance strategy. In this paper we look into angular rather than translational state variables since they are readily available for measurement and are less error-prone. Furthermore, the decision surfaces that we develop throughout this paper are based on state space models because they provide insight on controllability and stability of the system.

The procedure to determine the decision surface is to first consider an initial condition that drives the system to the border of stability. At this state the CoP reaches the edge of the foot and $T_a = -mg\delta$, where δ is the distance from the ankle joint to the edge of the foot (Fig. 1). Subbing this value for T_a into the state space representation of (2) yields

$$\begin{aligned} \dot{X} &= \mathbf{A}X + \mathbf{B}U \\ \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ \omega^2 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ -\omega^2 \end{bmatrix} f, \end{aligned} \quad (3)$$

where $f = \delta/L$. The initial values of θ and $\dot{\theta}$ for which the CoP reaches the edge of the foot delineate the borders of our decision surface. Taking the Laplace transform of (3), and rearranging yields

$$X(s) = \Phi(s)X(0) + \Phi(s)\frac{\mathbf{B}f}{s}, \quad (4)$$

where $\Phi(s) = \text{inv}(s\mathbf{I} - \mathbf{A}) = \frac{1}{s^2 - \omega^2} \begin{bmatrix} s & 1 \\ \omega^2 & s \end{bmatrix}$. Taking the inverse Laplace then yields

$$X(t) = \Phi(t)X(0) + \int_0^t \Phi(t - \tau)\mathbf{B}f d\tau, \quad (5)$$

where $\Phi(t) = \begin{bmatrix} \cosh(\omega t) & \frac{1}{\omega} \sinh \omega t \\ \omega \sinh \omega t & \cosh(\omega t) \end{bmatrix}$ and

$$\int_0^t \Phi(t-\tau) \mathbf{B} f d\tau = \begin{bmatrix} \frac{1}{\omega} \sinh \omega t & -\frac{1}{\omega^2} + \frac{1}{\omega^2} \cosh \omega t \\ -1 + \cosh(\omega t) & \frac{1}{\omega} \sinh \omega t \end{bmatrix} \begin{bmatrix} 0 \\ -\omega^2 f \end{bmatrix}$$

Replacing in (5) we get

$$\begin{aligned} \theta(t) &= (\theta_0 - f) \cosh(\omega t) + \frac{\dot{\theta}_0}{\omega} \sinh \omega t + f \\ &= \frac{1}{2}(\theta_0 + \frac{\dot{\theta}_0}{\omega} - f)e^{\omega t} + \frac{1}{2}(\theta_0 - \frac{\dot{\theta}_0}{\omega} - f)e^{-\omega t} + f \end{aligned} \quad (6)$$

Note that $\mathbf{A}$ has a positive eigenvalue ω that causes system instability. Here we will resort to making this mode unobservable in the output by forcing its coefficient to zero such that

$$\theta_0 + \frac{\dot{\theta}_0}{\omega} = f \quad (7)$$

Given that the foot has a front and rear edge the decision surface can then be defined using $f^\pm = \delta^\pm/L$ as

$$f^- < \theta_0 + \frac{\dot{\theta}_0}{\omega} < f^+ \quad (8)$$

Anytime θ and $\dot{\theta}$ of the system are such that (8) does not hold the ankle strategy is no longer capable of preventing the robot from falling and an alternate solution such as a momentum strategy is required.

2.2. Momentum strategy

A well studied human reflex is the momentum strategy, where a human rotates his/her upper body parts (torso and arms) in the direction of movement in order to create a momentum to counteract the applied disturbance. Similar to the inverted pendulum model, Pratt^{3,6,7} discussed the use of the Angular Momentum Pendulum Model (AMPM) to represent this type of reflex. The AMPM consists of replacing the point mass in the inverted pendulum model by a flywheel in order to include a momentum source (see Fig. 2). A flywheel is typically used for satellite orientation and other applications, and although it is beneficial to include a flywheel inside a real humanoid robot, to our knowledge this has never been attempted. Nevertheless, with appropriate mapping between the flywheel model and the humanoid model all the analysis performed on the flywheel can be applied for the humanoid. For the model proposed in this paper the robot adopts a momentum strategy each time the states

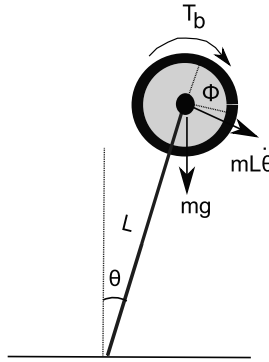


Fig. 2. The angular momentum pendulum model.

reach beyond the limits of the ankle strategy decision surface. Equations of motion for this model are

$$mL^2\ddot{\theta} = mgL\sin\theta - T_b \quad (9)$$

$$T_b = -I\ddot{\phi}, \quad (10)$$

where I is the moment of inertia of the flywheel. By linearizing the gravity term for small θ it can be considered that $\sin(\theta) \simeq \theta$ and (9) in state space representation becomes

$$\begin{aligned} \dot{X} &= \mathbf{A}X + \mathbf{B}T_b \\ \begin{bmatrix} \dot{\theta} \\ \dot{\phi} \\ \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \omega^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \phi \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1/mL^2 \\ -1/I \end{bmatrix} T_b \end{aligned} \quad (11)$$

Accordingly, the robot should be able to compensate for any force by applying an appropriately sized torque and with the right control can always be brought back to equilibrium state. In reality this is not true because the actuator saturates and the flywheel cannot rotate indefinitely; hence the need for a decision surface similar to that of the ankle strategy. However, instead of using the CoP location as the limiting factor as was done for the ankle strategy, here the limiting factors are the maximum torque and maximum angle that the flywheel can rotate through; we denote these values as T_{bm} and ϕ_m respectively. Let us first frame the problem as that of initial value problem in which the initial conditions are chosen to bring the robot to the limits of the momentum decision surface. At that point the flywheel exerts a maximum torque in order to bring CoM to its equilibrium point, followed by a torque in the opposite direction in order to stop rotation of the flywheel. In the extreme case

(the edge of the decision surface) the torque is

$$T_b = T_{bm}u(t) - 2T_{bm}u(t - t_1) + T_{bm}u(t - t_2), \quad (12)$$

where $u(t - \zeta)$ is the unit step function starting at ζ , t_1 is the time at which the flywheel stops accelerating and starts decelerating and t_2 is the time at which the flywheel comes to a stop. At t_2 we want $\dot{\phi}(t_2) = 0$ and $\phi(t_2) = \phi_m$. For constant accelerations and decelerations $t_2 = 2t_1$ and $t_1 = \sqrt{\frac{\phi_m I}{T_{bm}}}$. The solution for the model in (11) is of the form

$$X(t) = \Phi(t)X(0) + \int_0^t \Phi(t - \tau)\mathbf{B}T_b d\tau,$$

where Φ is the state transition matrix. Replacing T_b by its value in (12), at time $2t_1$ we get

$$\begin{aligned} X(2t_1) = & \Phi(2t_1)X(0) + \int_0^{t_1} \Phi(2t_1 - \tau)\mathbf{B}T_{bm} d\tau \\ & - \int_{t_1}^{2t_1} \Phi(2t_1 - \tau)\mathbf{B}T_{bm} d\tau \end{aligned} \quad (13)$$

The solution of (13) can be shown to be

$$\begin{aligned} \theta(2t_1) = & \cosh(2\omega t_1)\theta_0 + \frac{\dot{\theta}_0}{\omega} \sinh(2\omega t_1) \\ & - \frac{T_{bm}}{mL^2\omega^2} [\cosh(2\omega t_1) - 2\cosh(\omega t_1) + 1] \\ \dot{\theta}(2t_1) = & \omega \sinh(2\omega t_1)\theta_0 + \dot{\theta}_0 \cosh(2\omega t_1) \\ & - \frac{T_{bm}}{mL^2\omega} [\sinh(2\omega t_1) - 2\sinh(\omega t_1)] \end{aligned} \quad (14)$$

The role of the Momentum reflex is to bring back the state of the humanoid into the ankle strategy decision surface so that ankle strategy can be implemented. To satisfy this condition, the condition in (8) should be valid at time $t = 2t_1$

$$f^- < \theta(2t_1) + \frac{\dot{\theta}(2t_1)}{\omega} < f^+ \quad (15)$$

Replacing (14) in (15) and dropping the subscript for t_1 the decision surface for momentum strategy becomes

$$f^- < e^{2\omega t}\theta_0 + \frac{e^{2\omega t}}{\omega}\dot{\theta}_0 - \frac{T_{bm}}{mL^2\omega^2} [e^{2\omega t} - 2e^{\omega t} + 1] < f^+ \quad (16)$$

At anytime, when θ and $\dot{\theta}$ are such that the robot state is located outside the decision surface delineated by 16 the momentum strategy will not suffice. Stepping strategy will be necessary to bring back the system into the momentum strategy and eventually to the ankle strategy. The topic of stepping strategy will be addressed in future work.

After looking into a decision surface for both ankle and hip strategies, an appropriate feedback control scheme is required to ensure that the robot is appropriately conditioned to these constraints.

3. Control

In what follows, a 3DoF biped robot is considered and is shown in Fig. 3. It is composed of an unactuated DoF at the foot-ground interaction and ankle and hip actuated joints. The analysis is simplified by considering the motion of the robot only in the sagittal plane. The analysis of the system is simplified by disregarding the mass of the legs of the robot in the calculation of the center of mass. This assumption is valid for ankle and hip strategies given that the motion of the legs is minimal during the application of these strategies. For stepping strategy the legs would have to be included given that their motion could be substantial. Referring to Fig. 3(a) let β denote the angle of rotation of the foot as shown in and D denote the distance from foot base to the edge of the foot. β smaller than, equal to or larger than zero respectively indicates that D equals to δ^+ , 0, or δ^- . The 3DoF model is equipped with sensors that measure angular position and angular speed at each joint, as well as sensors that measure rotational displacement and velocity of the torso with respect to the world coordinates. Let $\gamma = \frac{\dot{\beta}}{2} + \dot{a} + \dot{h} + \dot{\beta}$ and $\dot{\gamma} = \ddot{a} + \ddot{h} + \ddot{\beta}$ represent the net angular motion and angular velocity of the torso of the simulated humanoid.

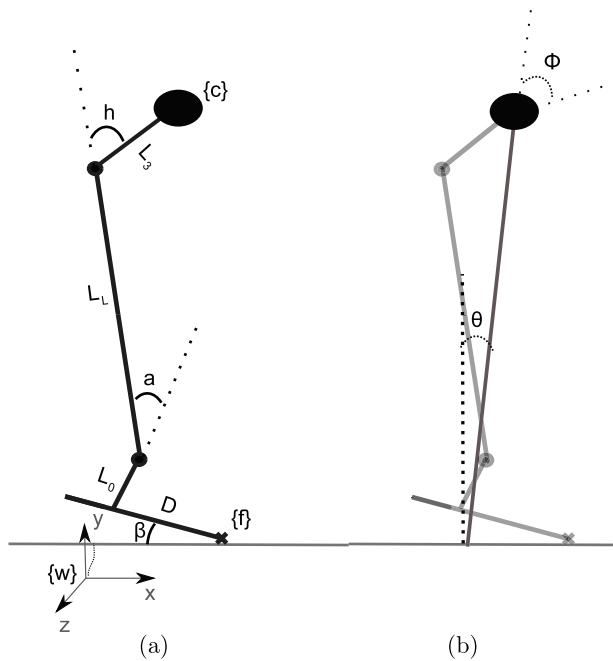


Fig. 3. The 3DoF model. (a) The 3DoF model (b) The equivalent inverted pendulum.

On the physical robot these values are measured by fixing a gyroscope and tilt sensor onto the robot. Notice in Fig. 3(a) that the world frame is located on the ground projection of the ankle joint. The transformation matrix from the world frame $\{w\}$ to the frame attached to the center of mass $\{c\}$ is found to be

$$\mathbf{T}_c^w = \begin{bmatrix} s_{\beta+a+h} & -c_{\beta+a+h} & 0 & D - (Dc_\beta + L_0s_\beta + L_Ls_{\beta+a} + L_3s_{\beta+a+h}) \\ -c_{\beta+a+h} & -s_{\beta+a+h} & 0 & -Ds_\beta + L_0c_\beta + L_Lc_{a+\beta} + L_3c_{\beta+a+h}) \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (17)$$

In Fig. 3(b) shown in red is the inverted pendulum simplification of the 3DoF robot. Normally, the base of such a pendulum is located at $D = \delta^+$, 0 or δ^- ; however, in order to be consistent with the previous analysis of the decision surfaces, it is necessary to consider the base at the $\{w\}$ origin. The forward kinematics for the humanoid robot yields $\theta = \text{atan}(\frac{T_c^w(1,4)}{T_c^w(2,4)})$ which can be linearized for small values of θ to $\theta \simeq \frac{T_c^w(1,4)}{T_c^w(2,4)}$, or

$$\theta = \frac{D - (Dc_\beta + L_0s_\beta + L_Ls_{\beta+a} + L_3s_{\beta+a+h})}{-Ds_\beta + L_0c_\beta + L_Lc_{a+\beta} + L_3c_{\beta+a+h}} \quad (18)$$

Note that $\dot{\theta}$ can be computed from (18) and the computed θ and $\dot{\theta}$ form the states of the 3DoF model, which are used in conjunction with the decision surfaces developed previously. In the coming section, a control law for the Ankle strategy of an inverted pendulum model is developed as well as the control law for the AMPM model.

3.1. Ankle strategy: Virtual model control

An interesting control algorithm that is put forward by Pratt *et al.*¹⁷ is the Virtual Model Control (VMC). VMC is a control framework that consists of connecting virtual physical components such as mechanical springs, dampers, and so on at appropriate locations on the robot. The interaction of these virtual components with the environment automatically creates the appropriate forces and torques at the actuators in order to perform controls corresponding to desired motions. The benefits of VMC are numerous including (1) the intuitiveness of the approach to control, (2) the ease in evaluating the effect of the control gains consisting of spring constants and damping ratios, (3) the lack of need for an exact dynamic model of a robot in our control algorithm, and (4) if appropriately used VMC can help linearize a system and subsequently allows one to apply all classical linear control methods. Since only the ankle strategy is considered here, the knee and hip joints are considered locked. Considering the model shown in Figs. 3 and 4(b) shows the virtual components in red. A spring and damper are connected on one side to the CoM and to a frictionless/weightless roller on the other. These virtual components produces appropriate horizontal forces F_s and F_d that bring the robot back to its initial equilibrium position. F_y is a vertical virtual force, which has the role of canceling the effect of

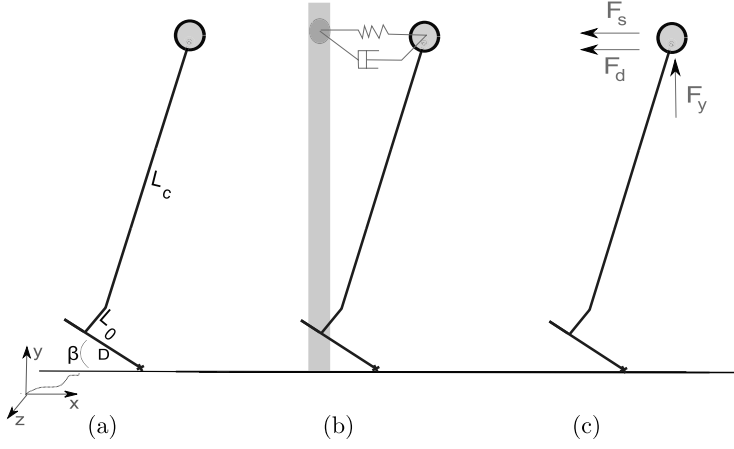


Fig. 4. Virtual model control applied on the inverted pendulum model. (a) Frame (b) virtual elements (c) virtual forces.

gravity ($F_y = mg$). Picking off the last column of the transformation matrix of (17) the position of CoM with respect to global coordinates is found to be

$$\mathbf{X}_c^W = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} D - (Dc_\beta + L_0s_\beta + L_cs_{a+\beta}) \\ -Ds_\beta + L_0c_\beta + L_cc_{a+\beta} \end{bmatrix}, \quad (19)$$

Partial differentiation with respect to 'a' produces the Jacobian

$$\mathbf{J} = \begin{bmatrix} \partial x / \partial a \\ \partial y / \partial a \end{bmatrix} = \begin{bmatrix} -L_cc_{a+\beta} \\ -L_cs_{a+\beta} \end{bmatrix} \quad (20)$$

This Jacobian relates the virtual forces to the joint torque as follows

$$\begin{aligned} \tau_a &= \mathbf{J}^T \mathbf{F} \\ &= \begin{bmatrix} -L_cc_{a+\beta} & -L_cs_{a+\beta} \end{bmatrix} \begin{bmatrix} F_s + F_d \\ F_y \end{bmatrix} \\ &= -L_c(c_{a+\beta})(F_s + F_d) - L_c(s_{a+\beta})F_y, \end{aligned} \quad (21)$$

where τ_a is the torque at the ankle joint. The forces in the spring, damper and y position can be expressed as

$$F_s = Kx = K[D - (Dc_\beta + L_0s_\beta + L_cs_{a+\beta})] \quad (22)$$

$$F_d = B\dot{x} = B[\dot{\beta}(Ds_\beta - L_0c_\beta) - (\dot{a} + \dot{\beta})L_c(c_{a+\beta})] \quad (23)$$

$$F_y = mg \quad (24)$$

Replacing in (3.1) yields the ankle torque for controlling the robot based on the VMC control formulation. One important ramification with VMC is that since the gains K and B represent spring and damper constants, it is possible to get physical insight into the way they are affecting the system. Also, because gravity is removed and the

system is that of a linear spring and damper the model can be analyzed using traditional linear system control and gains can be found using methods like pole placement and Linear Quadratic Regulator (LQR).

A numerical simulation was performed on an inverted pendulum model with maximum and minimum CoP distances constrained by saturation torques that were enforced onto the foot. The physical characteristics of the inverted pendulum simulation are shown in Table 1 and the phase plots of the numerical simulations are shown in Fig. 5. Note that the ankle strategy decision surface limits are drawn in black and the green circles represent the location of the initial conditions. In this figure two sets of simulations are shown; the first are the blue trajectories, which consist of $\theta_0 = 10^\circ$ and $\dot{\theta}_0$ varying and the second are the red trajectories, which consist of θ_0 varying and $\dot{\theta}_0 = 150^\circ/\text{sec}$. Note how the simulation predicts on the graph that the robot would not be able to recover when the initial conditions place the robot state outside of the ankle strategy decision surface.

Table 1. Inverted pendulum characteristics.

l	0.215 m
m	2 kg
δ^+	-0.025 m
δ^-	0.075 m
VMC spring constant K	30
VMC damper constant B	20

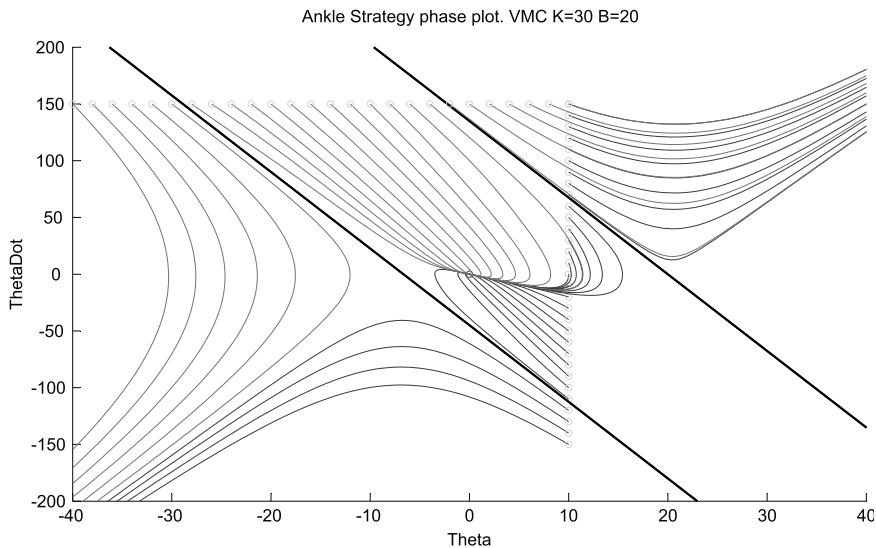


Fig. 5. Phase plots of the numerical simulation on the inverted pendulum model.

3.2. Momentum strategy PD controller

Recall that the control of the AMPM model, represented by (11) and shown in Fig. 2 involves applying a flywheel torque that rotates the flywheel in the falling direction in order to recover from a push disturbance, but also involves stopping the flywheel once the AMPM stabilizes. A control law that represents this behavior well is a Proportional-Derivative (PD) in place of the desired added momentum, and a derivative (D) controller that damps the flywheel rotation

$$T_b = \mathbf{F}\mathbf{X} \tag{25}$$

$$\mathbf{F} = \begin{bmatrix} K_f & 0 & B_f & -V_f \end{bmatrix} \tag{26}$$

$$T_b = K_f\theta + B_f\dot{\theta} - V_f\dot{\phi}, \tag{27}$$

where K_f , B_f and V_f are the respective gains of the controller. Optimized gains can be found using LQR. As in the previous section, a numerical simulation was also performed here on the AMPM model with saturation torque and saturation angle on the flywheel. The model is the one shown in Fig. 2 and the physical characteristic of the system are shown in Table 2. The resulting phase plots of the numerical simulations are shown in Fig. 6, where the Momentum strategy decision surface limits are

Table 2. AMPM model characteristics.

flywheel maximum angle ϕ_m	90°
flywheel maximum torque T_{bm}	1.617 N.m
flywheel moment of inertia I	0.027 kg.m ²
Controller K_f	63.2
Controller B_f	9.4
Controller V_f	1.3

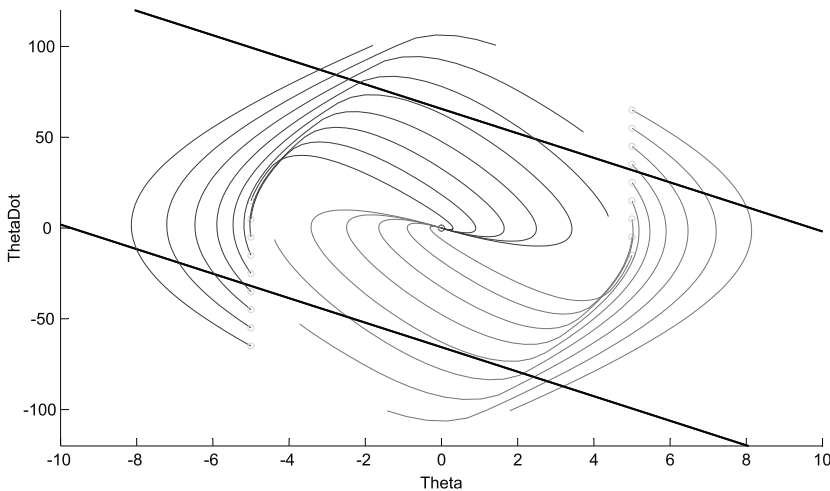


Fig. 6. Phase plots of the numerical simulation on the AMPM.

drawn in black and the green circles represent the initial conditions. Two sets of simulations are shown, the first are the blue trajectories that consist of $\theta_0 = -5^\circ$ and $\dot{\theta}_0$ varying and the second are the red trajectories that consist of $\theta_0 = 5^\circ$ and $\dot{\theta}_0$ varying. The simulation is run up until either the AMPM arrives at the equilibrium point or the flywheel reaches its maximum angle limit. For the trajectories starting outside the decision surface, the simulation stops before reaching the equilibrium point indicating that the AMPM is not able to recover and the flywheel saturates at its maximum angle. This behavior is shown in Fig. 7, where the flywheel angles quickly increase and saturate towards the maximum value. Figure 8 depicts how the

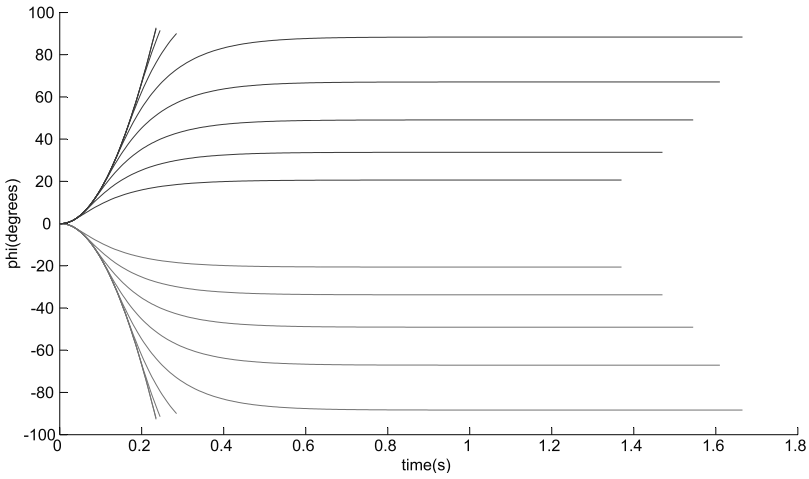


Fig. 7. Variation of the flywheel angle ϕ with time.

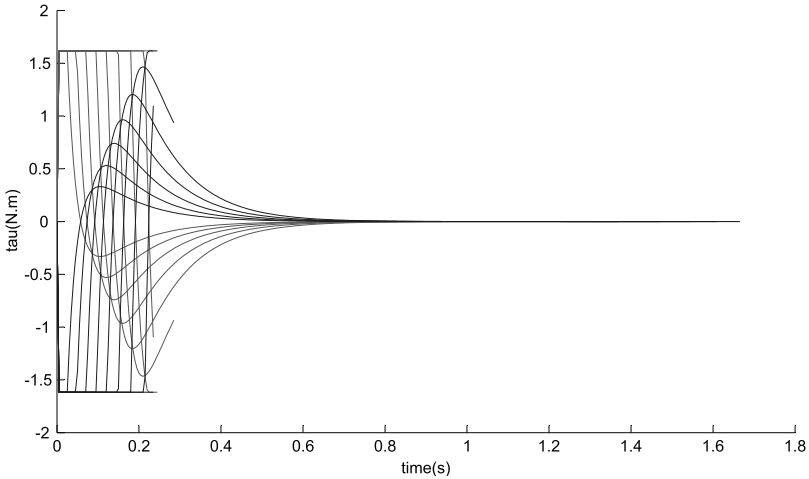


Fig. 8. Variation of the flywheel torque T_b with time.

flywheel torque varies during saturation, where it quickly increases and then decreases toward zero.

4. Application on the 3-DoF Model

A more realistic study of humanoid fall avoidance requires one to consider the robot as a 3DoF mechanism. In this work the ‘momentum’ strategy is more appropriately named ‘hip’ strategy since the former notation refers to a scheme where both the hips and arm are rotated to counteract a disturbance, while in our work to simplify the analysis only the hips are used to inject momentum into the system. In the case of the 3DoF model the CoM is located, contrary to the AMPM model, at an offset distance L_3 from the hip joint as can be seen in Fig. 9. Due to this offset, any rotation of the hips causes a motion of the CoM that might drive the system to instability. It is for this reason that the hip strategy can not be applied independently of the ankle strategy; instead, for strong push disturbances the two strategies should work together and at the same time. Note that in Fig. 9 τ_f is the torque control for the hip strategy, while F_x and F_y are the ankle strategy VMC forces. The transformation matrix of $\{c\}$ (located at the CoM) with respect to $\{w\}$ (world frame) is found in equation (17). Then the posture vector of frame $\{c\}$ with respect to world frame $\{w\}$ is

$$\mathbf{X}_c^w = \begin{bmatrix} x \\ y\gamma \end{bmatrix} = \begin{bmatrix} D - (Dc_\beta + L_0s_\beta + L_Ls_{\beta+a} + L_3s_{\beta+a+h}) \\ -Ds_\beta + L_0c_\beta + L_Lc_{a+\beta} + L_3c_{\beta+a+h} \\ \beta + a + h \end{bmatrix}, \quad (28)$$

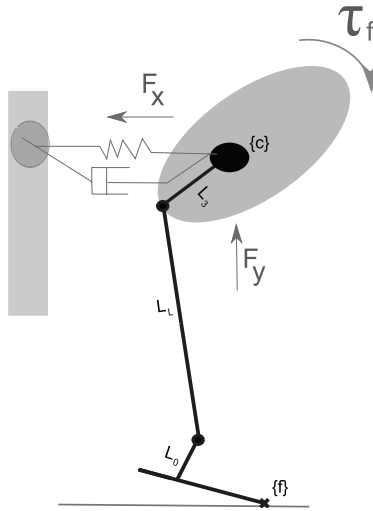


Fig. 9. VMC on the 3DoF biped model. The control forces and torque are in red.

with γ being the total rotation of the center of mass with respect to the world frame. This results in a Jacobian of the form

$$\mathbf{J}_{a^c} = \begin{bmatrix} \partial x / \partial a & \partial x / \partial h \\ \partial y / \partial a & \partial y / \partial h \\ \partial \gamma / \partial a & \partial \gamma / \partial h \end{bmatrix} = \begin{bmatrix} -L_L c_{\beta+a} - L_3 c_{\beta+a+h} & -L_3 c_{\beta+a+h} \\ -L_L s_{\beta+a} - L_3 s_{\beta+a+h} & -L_3 s_{\beta+a+h} \\ 1 & 1 \end{bmatrix} \quad (29)$$

and the torque can subsequently be expressed as

$$\begin{aligned} \tau &= \mathbf{J}_{a^c}^T F \\ \begin{bmatrix} \tau_a \\ \tau_h \end{bmatrix} &= \begin{bmatrix} -L_L c_{\beta+a} - L_3 c_{\beta+a+h} & -L_L s_{\beta+a} - L_3 s_{\beta+a+h} & 1 \\ -L_3 c_{\beta+a+h} & -L_3 s_{\beta+a+h} & 1 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \\ \tau_f \end{bmatrix}, \end{aligned} \quad (30)$$

According to (28)

$$\begin{aligned} x &= D - (Dc_\beta + L_0 s_\beta + L_L s_{\beta+a} + L_3 s_{\beta+a+h}) \\ \dot{x} &= \dot{\beta}(Ds_\beta - L_0 c_\beta) - (\dot{a} + \dot{\beta})L_L(c_{a+\beta}) - (\dot{h} + \dot{a} + \dot{\beta})L_3(c_{\beta+a+h}) \end{aligned} \quad (31)$$

hence the virtual forces that are applied are

$$\begin{aligned} F_x &= K(D - (Dc_\beta + L_0 s_\beta + L_L s_{\beta+a} + L_3 s_{\beta+a+h})) \\ &\quad + B(\dot{\beta}(Ds_\beta - L_0 c_\beta) - (\dot{a} + \dot{\beta})L_L(c_{a+\beta}) - (\dot{h} + \dot{a} + \dot{\beta})L_3(c_{\beta+a+h})) \end{aligned} \quad (32)$$

$$F_y = mg, \quad (33)$$

where K and B are the virtual spring and damper gains that we need to tune. As for the momentum strategy and similarly to (25),

$$\tau_f = K_f \theta + B_f \dot{\theta} - V_f \dot{\beta}, \quad (34)$$

with θ being the angle of the pendulum as shown in Fig. 3 and defined in (18). As a control design criterion, it is requested that F_x affects only τ_a and τ_f affects only τ_h . The controller equations become

$$\tau_a = -(L_L c_{\beta+a} + L_3 c_{\beta+a+h})F_x - (L_L s_{\beta+a} + L_3 s_{\beta+a+h})F_y \quad (35)$$

$$\tau_h = \tau_f - L_3 s_{\beta+a+h}F_y \quad (36)$$

Note that τ_a is applied when the robot's states are located inside the ankle strategy decision surface, and the hip joint is locked. As soon as the robot detects that it will reach the decision surface boundary for the ankle strategy, the system adopts the hip strategy and τ_a and τ_h defined in (35) are applied. This strategy is displayed in the flowchart of Fig. 10.

It is important to note that although a combined strategy is adopted here, the decision surfaces that were developed in the previous sections still apply. Therefore, using the states θ and $\dot{\theta}$ that are computed from (18) the system can be approximated by an inverted pendulum and the same ankle strategy decision surface defined in (8) is applicable. As for the hip decision surface, we consider in this case the limiting

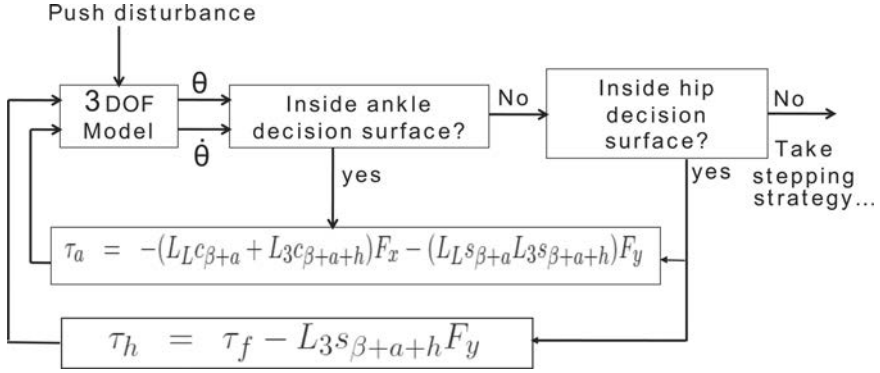


Fig. 10. A flowchart showing the ankle and hip strategy control system.

factors to be the maximum torques and angles of the hip τ_{hm} and β_m rather than those of the flywheel. Again using the states θ and $\dot{\theta}$ computed from (18) the system can be approximated by an AMPM and the same momentum strategy decision surface defined in (16) can be applied.

5. Experiments and Results

This section is divided into two subsections, the first summarizes the experiments performed on the simulated robot, while the second outlines those performed on the real Bioloid robot.

5.1. Gazebo simulation

Push recovery simulations were performed on a realistic humanoid robot model. It was decided to use Gazebo simulator that is part of the player project.¹² Gazebo is a 3D simulator with the dynamics provided by the Open Dynamics Engine (ODE).¹⁸ A humanoid model named 'BAUB' (Biped of the American University of Beirut) that was created under Blender was imported to Gazebo for this purpose. A robotic arm with a controllable speed of impact is used for the push experiments. The preference of having a robotic arm with different speeds of impact rather than having a controlled impulse force is to simulate a real life environment where a robot can be pushed with a random force. The physical characteristics of BAUB are shown in Table 3 where the naming standards that are used are taken from Fig. 3.

Figure 11 shows BAUB with the ankles and hip locked (no reflex). The robotic arm applied a push with a speed 105°/sec. BAUB recovers after heavily oscillating. Figure 12 shows BAUB undergoing VMC controlled ankle strategy to recover from the 105°/sec robotic arm push. Notice how in the first set of pictures the robot swings back and forth like a pendulum (Fig. 11f) while in the second set of pictures the ankle responds, even though the foot starts rotating on its front edge, and a smooth reaction occurs to the disturbance. In the latter case, the robot stops the forward fall

Table 3. The physical characteristics of BAUB.

Leg DoF	5
sensors	IMU + FSRs on bottom of feet
$L_0; L_L; L_3$	0.1 m; 0.75 m; 0.3 m
$\delta^+; \delta^-$	0.2 m; -0.15 m
Torso Moment of inertia I; Mass	0.2 kg.m ² ; 20 kg
Hip joint maximum angle $(\theta_h)_{\max}$	140°
Hip joint maximum torque $(\tau_h)_{\max}$	10 N.m

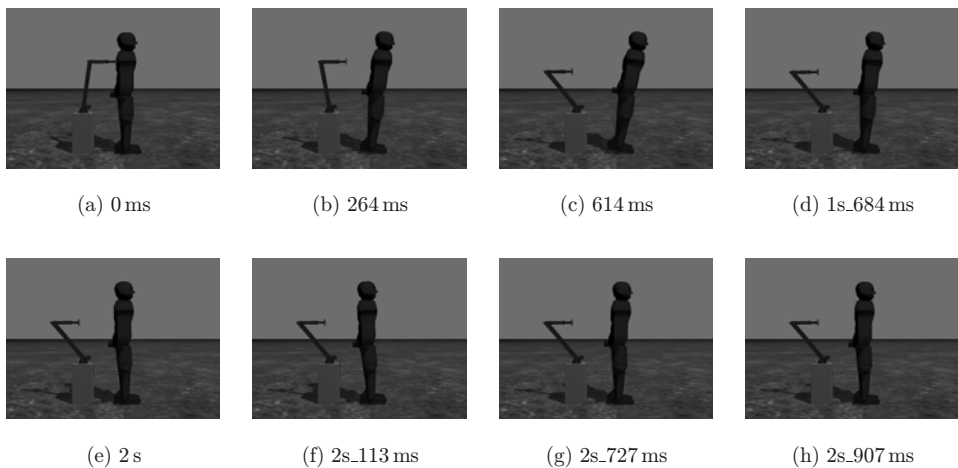


Fig. 11. Push with a speed of $105^\circ/\text{sec}$, ankles locked and reflex turned off.

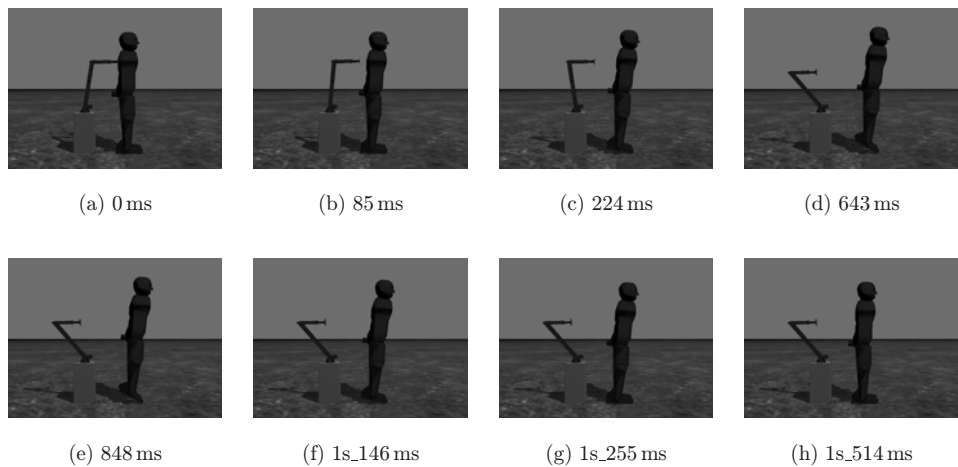


Fig. 12. Push with a speed of $105^\circ/\text{sec}$ reflex turned on.

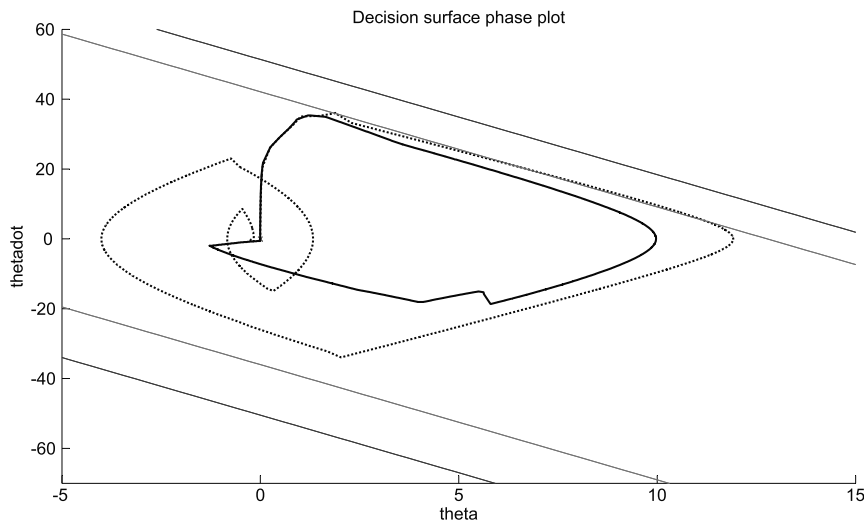


Fig. 13. Phase plot of BAUB push with a speed of $105^{\circ}/\text{sec}$.

while it is standing on its tip-toes. Figure 13 compares the two phase plot trajectories of ‘with’ and ‘without reflex’, where we notice that in the case of ‘without reflex’ there is bigger disturbance and oscillations while with VMC control the oscillations are damped. In both cases, the states remain inside the ankle strategy decision surface.

Figure 14 shows a push with $106^{\circ}/\text{sec}$ speed for the robotic arm, while the ankles and hips are locked. BAUB falls to the ground. The trajectory of this push in the phase plane is shown as the dotted line in Fig. 17. It is noted that the states reached beyond the ankle decision surface, hence hip strategy is required. Figure 15 shows the

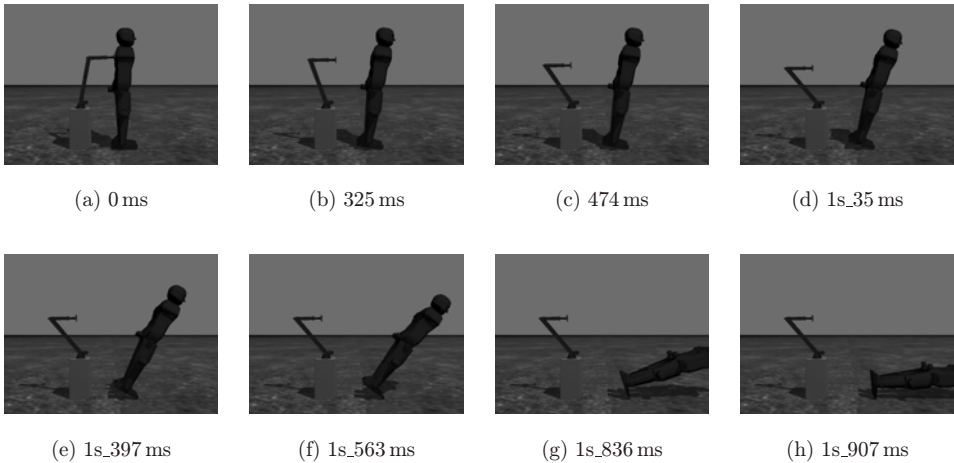


Fig. 14. Push with a speed of $106^{\circ}/\text{sec}$, ankles locked and reflex turned off.

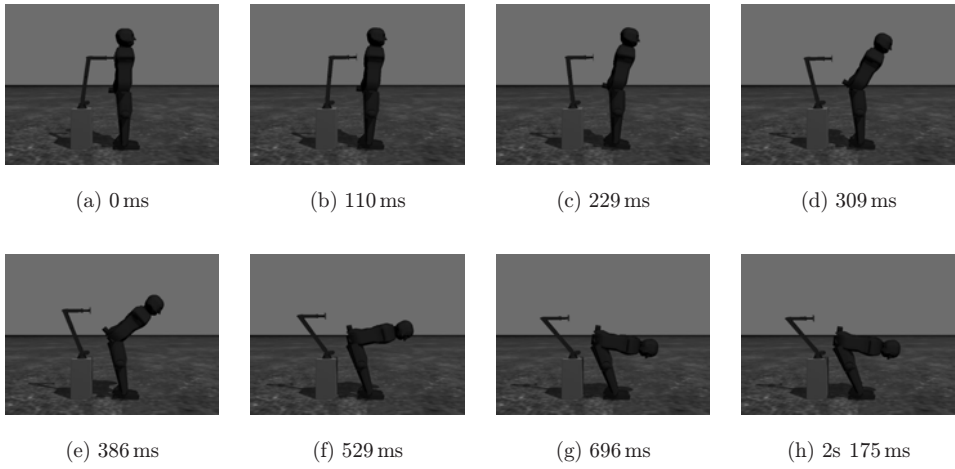


Fig. 15. Push with a speed of $106^\circ/\text{sec}$, reflex turned on.

same push but reflex turned on and BAUB uses both hip and ankle strategies simultaneously in order to recover. At $t = 230$ ms BAUB realizes that ankle strategy is not enough and the rotation of the hips is initiated. Figure 17 compares the two phase plot trajectories, where it is observed that without reflex the robot is unable to recover while with reflex the states reach the hip strategy decision surface and BAUB recovers using hip strategy.

Figure 16 shows BAUB undergoing a combined ankle and hip strategy to recover from a $109^\circ/\text{sec}$ robotic arm speed push. Comparing this figure to that of Fig. 15, it is noted that the hips reached the maximum angle allowed in order to recover, where BAUB reaches the edge of the hip strategy decision surface. Figure 18 compares the phase plots of disturbances of increasing speeds from 107 to $110^\circ/\text{sec}$, and what is

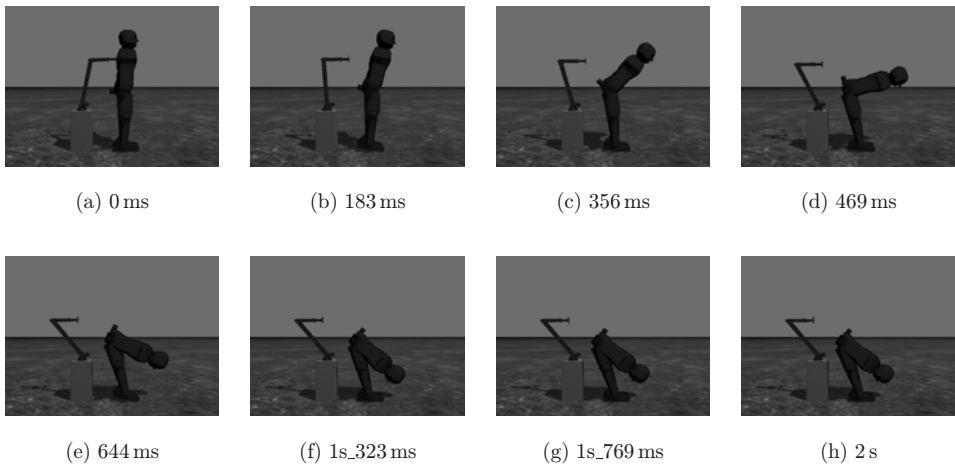


Fig. 16. Push with a speed of $109^\circ/\text{sec}$, reflex turned on.

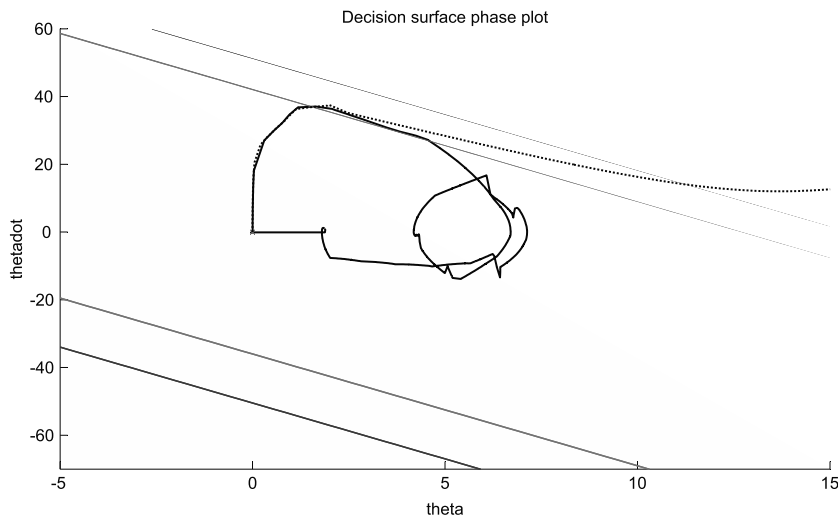


Fig. 17. Phase plot of BAUB push with a speed of $106^{\circ}/\text{sec}$.

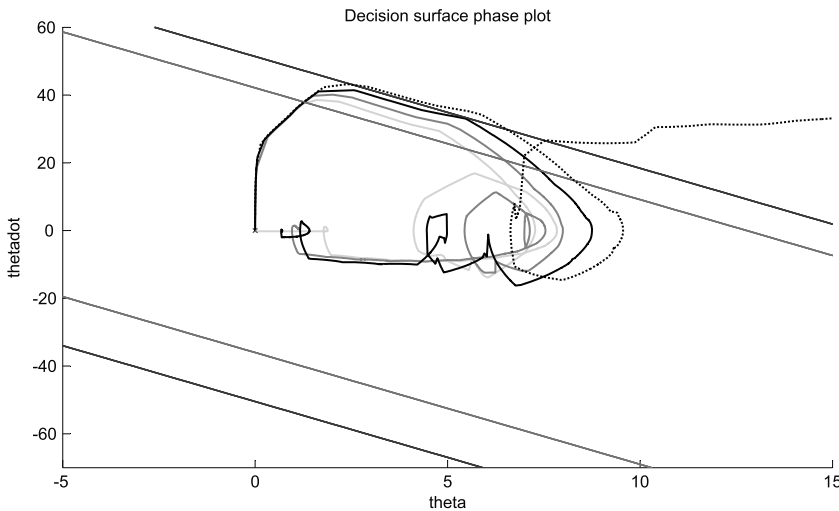


Fig. 18. Phase plot of BAUB push with speeds of $107, 108, 109$ and $110^{\circ}/\text{sec}$.

observed is that for $110^{\circ}/\text{sec}$ the location of the state moves beyond the hip decision surface limits and BAUB is not able to recover.

5.2. Bioloid robot

The theory of the hip and ankle strategy were implemented on a real humanoid robot. The robot used is named Bioloid, which is a small sized embedded system robot

manufactured by Robotis.¹⁰ This robot is retrofitted with Force Sensitive Resistors (FSR) force sensors on each foot in order to measure the location of the CoP, as well as a gyroscope sensor connected at the torso that measures the angular rate of rotational velocity $\dot{\gamma}$ of the torso and the angular displacement γ is subsequently obtained by integrating the gyroscope reading. Unfortunately, this integration is inaccurate and error prone due to the inherent drift generated when integrating the gyroscope. Therefore, a certain correction is required, which is done by using the location of the CoP as a guide. When the position of the CoP is not on the edge, the foot is flat on the ground, $\beta = 0$ and γ can be simply measured using the potentiometer readings $\gamma = a + h$. It is at that moment that the gyroscope integration can be corrected. By connecting force sensors on the foot soles of the Bioloid robot, the CoP position is measured. Using this reading, a Gain Varying Kalman Filter is used to fuse the potentiometer measurement and the gyroscope integration as can be seen in the flowchart of Fig. 19. This filter is useful in push recovery applications since the successful disturbance does not normally last more than hundreds of millisecond. Therefore, the gyro cannot drift for a significant amount of time before it is corrected by the potentiometer. The Kalman filter is a simple, one variable filter. Let Q be the process noise covariance (the covariance of the gyroscope integration) and let R be the measurement noise covariance (covariance of the potentiometer reading of γ). Let γ_k be the Kalman measured angle and γ_{fk} be the potentiometer (forward kinematics) measured angle. The predicted values of γ and of P (the error covariance) are

$$\gamma_k^- = \gamma_{k-1} + (\dot{\gamma})dt \quad (37)$$

$$P_k^- = P_{k-1} + Q \quad (38)$$

The calculated Kalman gain is

$$K_k = \frac{P_k^-}{P_k^- + R} \quad (39)$$

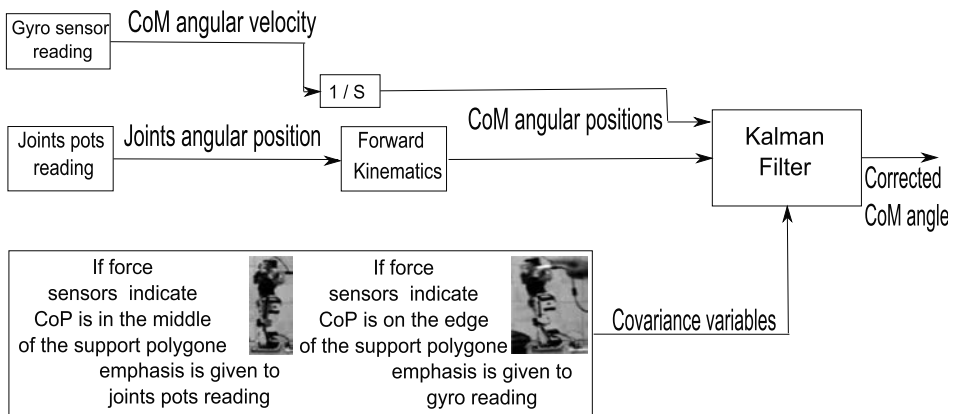


Fig. 19. Gain Varying Kalman filter.

Then the corrected angular position is

$$\gamma_k = \gamma_k^- + K_k(\gamma_{fk} - \gamma_k^-) \quad (40)$$

$$P_k = (1 - K_k)P_k^- \quad (41)$$

If CoP is inside the foot soles, the measurement covariance is set to be low while the process covariance is set relatively high such that the gain is close to 1 and more emphasis is given to the potentiometer measurement. If CoP is close to the foot edge the contrary is required.

As can be seen in Fig. 20, in order to test the gain varying Kalman filter, the robot was subjected to a push disturbance while the ankle and hip joints are locked (the foot is allowed to rotate on the ground). It is noticed in Fig. 20 that while the CoP is inside the foot soles, more emphasis is given to the pot readings, and the Kalman filter is not affected by the gyroscope's obvious drift. For the few milliseconds that the CoP is outside the foot soles, more emphasis is given to the gyro, and the Kalman filter tracks the gyro's trend while the potentiometer remains in the zero position.

Note that the controller developed in the previous section is a torque controller. The Bioloid robot does not provide torque control to its AX-12 actuators. For this reason, the closed loop speed control of the Bioloid robot was turned off and the PWM input was used to provide the torque control. To do so, an open loop correlation was found between the input Duty Cycle and the actuator output torque.

Figure 21 shows the ankle strategy after applying a moderate push on the Bioloid robot. It is noticed how the oscillations are damped with minimum rotation of the foot. Figure 22 shows the hip strategy being performed in combination with the ankle strategy. Both figures indicate the successful recovery of the Bioloid robot. The phase plots for both experiments can be seen in Fig. 23. The red line is the ankle decision surface while the blue line is that of the hip strategy. The plot on the left shows that the robot states remain inside the ankle strategy decision surface, hence ankle

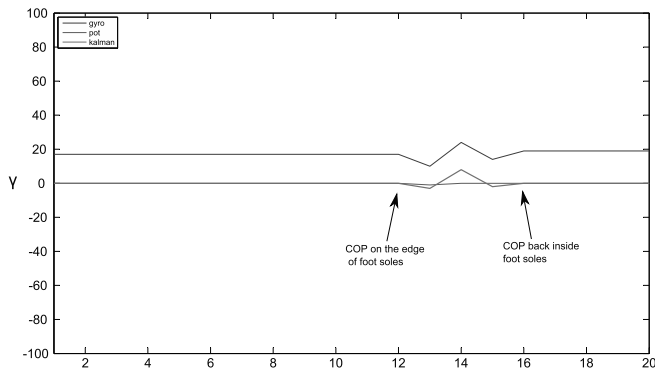


Fig. 20. Comparison between the γ reading of the gyroscope, the pots and the Kalman filter.

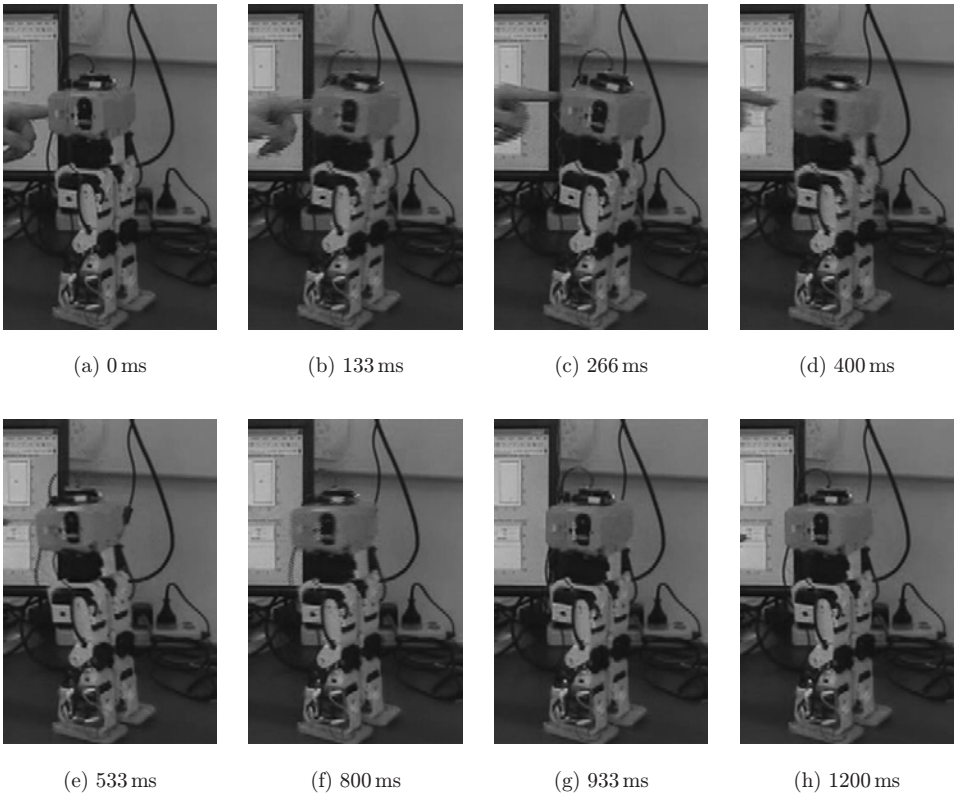


Fig. 21. Ankle strategy on Bioloid robot.

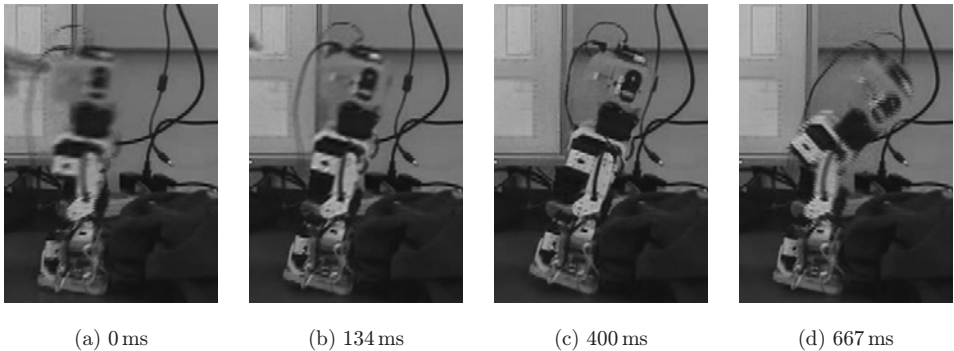


Fig. 22. Hip strategy on Bioloid robot.

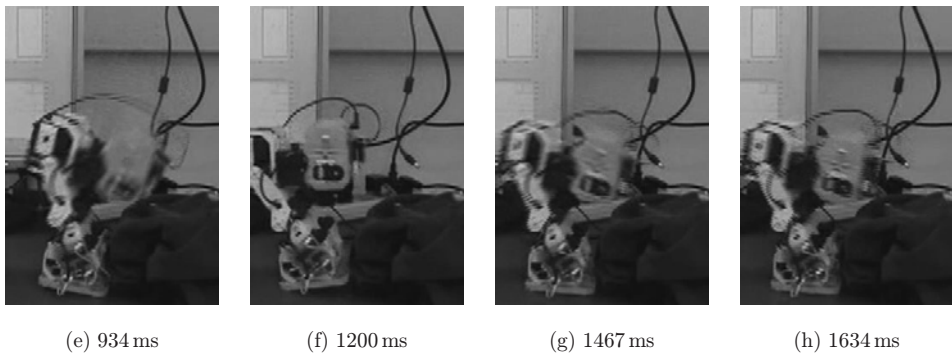


Fig. 22. (Continued)

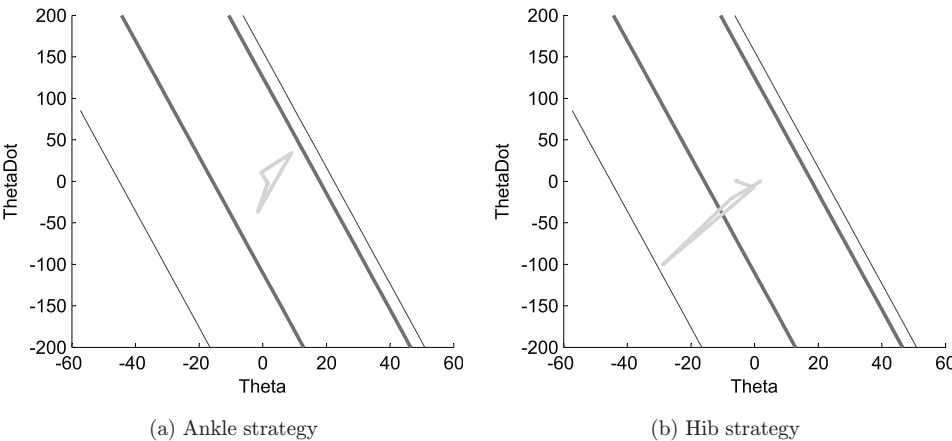


Fig. 23. Phase plots for the bio-inspired robot disturbances.

strategy is used to successfully recover. As for the plot on the right, the states went beyond the ankle strategy limits and hip strategy was initiated in order to recover.

6. Limitation, Conclusions and Future Work

6.1. Limitations

In this work, the push recovery problem of bipedal robots is solved in the 2D sagittal plane. Moreover, the robot is able to recover from disturbances occurring on the perpendicular to that plane only. If successful push recovery is to be implemented for humanoid robots, the robot should be capable of recovering for disturbances occurring on all sides, and the theory needs to be implemented on a 3D model. To do so, the robot needs to be considered as consisting of an inverted pendulum with two supports instead of one and having three dimensional feet on the ground.

On another note, during the entire development, the CoM was considered as having a fixed position in space (located on the torso), this is due to the assumption that the legs are massless. In the sagittal plane, this assumption can be considered acceptable, since the rotation of the legs are limited relative to the torso, and the torso is usually the heaviest link on a humanoid body. However, while working on the simulated and the real robot, this assumption as well as other linearizing assumptions that were used affected the precision of the hip strategy decision surface, which subsequently required some manual empirical fine tuning in order to improve its precision in making the correct decision. Machine learning algorithms could have been used to fine tune these decision surface when applied on the real robot. Moreover, if push recovery is to be implemented on a 3D humanoid robot, the mass of different body links (i.e., the arms and the legs) could be used in order to intentionally move the CoM position in space for the purpose of balancing. This consists of a new reflex strategy that needs to be investigated.

Finally, the results that were obtained on the Bioloid robot were not as impressive as the ones found in the simulations. The reason for this is the open loop torque control that was applied by the actuators. If the theory is to be implemented on a humanoid robot, it is preferable that this robot uses closed loop feedback torque control. Such closed loop control could be achieved using a method such as Series Elastic Actuators.²²

6.2. Conclusions and future work

This paper presented a fall avoidance system for humanoid robots that is capable of choosing amongst different reactive strategies including ankle and hip fall avoidance schemes. A decision surface was developed for both schemes and results show that these surfaces successfully predict the boundaries beyond which each of the fall avoidance schemes fails. During the tests our simulated humanoid was pushed with disturbances of different magnitudes and it consistently chose the correct reaction to take. The work developed in this paper was performed on a 2D model with disturbances happening in the sagittal plane. Our future work includes importing the theory into a 3D robot by developing decision spaces instead of decision surfaces. Since many model simplifications were used in the current development of the decision surfaces, machine learning could be used in order to fine tune these decisions surfaces for the real humanoid robot. Currently we are in the process of porting the theory into a torque controlled humanoid robot. In the future work stepping strategy will be incorporated into the fall avoidance system.

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